

OLGA FRIDMAN

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SUMMARY

Highly experienced Software Development Engineer with over 12 years of expertise in designing, developing, and implementing software solutions. Proficient in **C#** (10+ years) and **WPF(XAML)** (8+ years), with a strong background in developing cloud-based applications, web services, and tools for diverse industries including technology, hardware, finance, and gaming. Adept at collaborating with cross-functional teams and delivering high-quality software.

Sr. Software Engineer at Microsoft

January 2025 -July 2025

Designed and implemented a cross-platform CLI tool in .NET 9 using Spectre.Console, delivering a custom, fault-tolerant dataset transfer pipeline optimized for very large AI training datasets, addressing reliability and scalability limitations encountered with existing azcopy-based workflows. Backend services built on .NET Aspire implemented Azure Resource Manager (ARM)-based APIs to audit and govern dataset transfer operations, ensuring traceability, policy enforcement, and secure access via Microsoft Entra ID (AAD). The system integrates with Azure Blob Storage for scalable storage and Azure Cosmos DB for durable metadata management and audit logging. This solution enabled reliable, efficient transfers at scale, significantly reduced manual intervention, and supported reproducible model development across MAI and Azure GenAI teams.

SW Principal Engineer at Dell

2021 – 2024

- Developed a cloud-based solution (**C#/.NET, Docker, JSON**) for order orchestration with event-driven notifications for downstream systems using Confluent Kafka library to produce and send events to Event Hub in Azure:
 - **Microservices Development:** Developed and deployed microservices in Docker containers to handle various order processing and orchestration aspects.
 - **Database Management:** Designed and managed **PostgreSQL (PGSQL)** databases to store and retrieve order information efficiently.
 - **API Development:** Developed RESTful APIs to facilitate communication between different services and components within the orchestration system.
 - **Unit Testing:** Implemented unit tests using Mock to ensure the reliability and correctness of the code.
 - **Tavern Testing:** Developed and executed comprehensive API tests using **Tavern**, ensuring the reliability and correctness of interactions between services.
- Designed, implemented, and tested an HTTP-client tool (**C#, WPF/XAML**) to simulate order creation and test web service API requests across various environments for quality assurance purposes.
- Engineered a cloud-based system for sending notifications via email using **Java** and **SendGrid** (Azure service), ensuring reliable and scalable communication.

SDE Surface, Microsoft

2019 – 2020

Designed and implemented an XML-driven Debug Framework, enabling firmware engineers to perform board-level debug operations. This solution eliminates Surface products' dependency on external debugging software by detecting the motherboard and debug card, and retrieving the appropriate set of commands. The framework is reusable for any new Surface computer releases, enhancing long-term maintainability and reducing development time for future products.

Bridged the gap between managed C# tooling and native C++ firmware components by reverse-engineering message formats and validating microcontroller behavior, enabling the Surface debug platform to replace legacy vendor tools fully.

- **API Development:** Created a set of APIs using a multithreading architecture and third-party software to interact with serial ports, facilitating communication with microcontrollers on devices under test and significantly improving debugging capabilities and efficiency. (**.NET Framework, C#**)
- **Encoding/Decoding Engine:** Developed an encoding/decoding engine to handle hardware messages, ensuring accurate data interpretation.
- **User Interface:** Designed and built a user-friendly interface using **WPF(XAML)**, providing engineers with intuitive

access to debugging tools and information. (**.NET Core, C#**)

- **Unit Testing:** Implemented comprehensive unit tests to ensure the reliability and robustness of the debug framework.
- **PowerShell Scripting:** Wrote PowerShell scripts to automate repetitive tasks and streamline the debugging process.
- **Documentation:** Authored detailed technical documentation for the framework, APIs, and scripts to support user adoption and ongoing maintenance.
- **Collaboration:** Worked closely with firmware engineers and other stakeholders to gather requirements, address issues, and refine the framework.

SDE Microsoft

2018 – 2019

Developed a web-based project that presents and filters information about software products, their components, and versions for different devices. Supported and enhanced the solution (mid-layer **C#** and back-end **Kusto**).

SDE Visa

2017 – 2018

Developed an extendable, flexible, efficient, and evolving services platform that provides:

- General validation for Windows OS platform-based verticals, including Security, Authentication, Access Control, Network, Transport Connectivity, Name Resolution, NT Services, Registry, and XML/XSD.
- Validation for IIS Web Server and SQL Server, focusing on configuration, current availability, and health status.

Intended usage: to provide a fast and consistent way of validating the availability and health status of testing environments, as well as customer-specific applications, and to equip QA and support engineering teams with the necessary information for efficient root cause analysis, targeted diagnosing, and troubleshooting, thus reducing investigation and fix-up time.

SDE Microsoft, Turn10 Studios

2014 – 2016

- Designed and developed in-house tools for validating assets for a car race simulator (Forza). Replaced the old PowerShell version with a flexible, configurable, and interactive solution using **WPF(XAML)** and **C#**. It allowed a significant reduction in the game build time.
- Implemented various validations, including **XML** and **MAX** validation using **3D MAX** scripts.

Freelance Software Design Engineer

2007 – 2013

- Developed a set of game applications for Windows Phone and WinRT. **Vectorform LLC**.
- Designed and implemented a transliteration web service for transcribing text between writing systems. The service supports the Microsoft Office Research Service protocol and has an autonomous web page interface.
- Microsoft Office bilingual dictionary service. Designed and implemented the translation dictionary code and an efficient binary format. The service resides on a client machine as a Microsoft Office Research provider.

SKILLS

- Programming Languages: **C#, WPF(XAML), XML/XSD, PowerShell, JavaScript, C++, Java**
- Technologies: **Cosmos DB, Azure, ARM, Docker, Rest API, PGSQL, JSON, MVVM, Tavern API Testing, Git, Kusto**
- Tools: **Confluent Kafka, Event Hub, SendGrid, MS Build, 3D MAX, MEF**

EDUCATION

Master of Science in Applied Mathematics and Computer Science.

Rostov State University (Southern Federal University), Russia

LANGUAGES

Russian, English, Hebrew, French